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Descriptive Statistics
Academic Year 2025-2026
(Time: 3 hours)

Courses Questions: (05 points)

1- What is the difference between a univariate statistical series and a bivariate statistical series
2- What are the components of a time serie
3- Define the term **"Index Numbers"**? What are the uses of index numbers in decision making?
4- Underline the difference between a discrete random variable and a continuous random variable
5- Name two discrete distributions and two continuous distributions

Exercise I: (06 points)

The following table gives the values of Consumption (Y) by a household in the city of Yaoundé according to annual income (X in thousands of FCFA)

Income (X)	42	65	66	82	86	130
Consumption(Y)	38	64	63	74	79	107

1- Calculate average income and average consumption
2- Determine the covariance between X and Y noted COV (X, Y) as well as the Variance of X, V(X).
3- Knowing that the regression line equation of consumption (Y) as a function of income (X) is of the form $Y = aX + b$, where $a = \frac{COV(X,Y)}{V(X)}$ and $b = \bar{Y} - a\bar{X}$, derive a and b.

Exercise II: (05 points)

In the following contingency table, X represents a country's trade balance and Y the exchange rate.

X \ Y	[0,10[	[10,20[	[20,30[
[0,4[	0	40	0
[4,8[	20	0	20

1- Determine the marginal distributions of X and Y
2- Calculate the means and variances of these marginals' distributions

Exercise III: (04 points)

Three items X, Y, and Z were selected for the calculation of the cost-of-living index in Bamenda. We have the following information on prices (in FCFA/kg) and quantities consumed in (Kg) for the Year 2023 and 2024.

Items	P_0 2023	Q_0	P_1 2024	Q_1
	Prices (in FCFA)	Quantity (in Kg)	Prices (in FCFA)	Quantity (in Kg)
X	13	7	16	5
Y	5	12	8	12
Z	15	9	13	10

From the following data compute Laspeyre's index number and Paasche's price index number.

$L = \frac{P_1 Q_0}{P_0 Q_0} \times 100$ / $P = \frac{P_1 Q_1}{P_0 Q_1} \times 100$

Exercise IV: (05 points)

The descriptive statistics scores of 20 learners randomly selected from the Catholic University of Central Africa are given below:

69, 70, 93, 53, 92, 85, 75, 70, 68, 88, 76, 70, 77, 85, 82, 82, 80, 96, 99, 83

1- Compute the mean, median, standard deviation, variance, coefficient of variation, and interquartile range
2- Using a graph paper, draw a box and whisker plot to represent the above data. Define clearly all parameters. Comment on the shape of the distribution
3- Using the findings obtained (question 2), state with reasons, which measure of central tendency you consider more appropriate for summarizing these data.

Dr. NGAMBOE